Agenda

Short review & Class exercise

Confidence intervals

- For the mean difference between two samples
- For the population proportion and the difference between two population proportions
- For the mean of normal random samples with small sample size and unknown variance

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CI - Normal random sample

$X_1, \ldots, X_n$: random sample

$X_1, \ldots, X_n$ ~ Normal with mean μ and standard deviation σ.

Assume that σ is known.

Two-sided CI: A 100(1- α)% CI for the mean μ of a normal population when σ is known is given by:

$$\left(\bar{x} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}}, \bar{x} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right)$$

One-sided CI: A 100(1- α)% bounds for the mean μ of a normal population when σ is known is given by:

LCB: $\bar{x} - z_{\alpha}\frac{\sigma}{\sqrt{n}}$ UCB: $\bar{x} + z_{\alpha}\frac{\sigma}{\sqrt{n}}$

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LCB: $\bar{x} - z_\alpha \frac{\sigma}{\sqrt{n}}$ UCB: $\bar{x} + z_\alpha \frac{\sigma}{\sqrt{n}}$

Class exercise

A cargo company aims to have a mean delivery time of **at most 5 hours** for local deliveries. They believe delivery times are normal, with a standard deviation (σ) 2.4. They recorded 10 delivery times as:

7.3, 3.3, 4.7, 6.8, 10.1, 5.7, 6.4, 4.2, 3.1, 8.9,

with a sample mean of $\bar{x}$=6.05. Here are the confidence lower and upper bounds for different confidence levels:

1-α	α	z_α	LB	UB
0.9	0.1	1.28	5.08	7.02
0.95	0.05	1.64	4.80	7.30
0.99	0.01	2.33	4.28	7.82

Do you think the cargo company achieves its aim at a confidence level of 95%? At 90% or 99%? Why or why not?

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CI - Random sample with large sample size

$X_1, \ldots, X_n$: random sample

$X_1, \ldots, X_n$ ~ General distribution

Assume that σ is unknown but the sample size is large.

Two-sided CI: A 100(1- α)% CI for the mean μ of an arbitrary population when n is large is given by:

$$\left(\bar{x} - z_{\alpha/2}\frac{s}{\sqrt{n}}, \bar{x} + z_{\alpha/2}\frac{s}{\sqrt{n}}\right)$$

One-sided CI: A 100(1- α)% bounds for the mean μ of an arbitrary population when n is large is given by:

LCB: $\bar{x} - z_\alpha \frac{s}{\sqrt{n}}$ UCB: $\bar{x} + z_\alpha \frac{s}{\sqrt{n}}$

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Main idea:

Collect a random sample

Determine the parameter of interest, say θ

Find a random variable, say W, that involves parameter θ and whose distribution you can specify

Use the distribution of W to find random lower and upper bounds on θ

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Parameter of interest: true mean, μ

- $X_1, \ldots, X_n$: A random sample with $E[X] = \mu$
- We are looking for a random variable $Y = h(X_1, \ldots, X_n; \mu)$ whose distribution can be specified.

	X is normal		X is not known to be normal		
	n<40	n≥40	n<30	30≤n<40	n≥40
σ: known	$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$	$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$	?	$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$	$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$
σ: unknown	*To be discussed today*	$Z = \frac{\bar{X} - \mu}{S/\sqrt{n}}$	?	*To be discussed today*	$Z = \frac{\bar{X} - \mu}{S/\sqrt{n}}$

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CIs around the true mean, μ

- $X_1, \ldots, X_n$: A random sample with $E[X] = \mu$
- We are looking for a random variable $Y=h(X_1,\ldots,X_n; \mu)$ whose distribution can be specified.

	X is normal		X is not known to be normal		
	n<40	n≥40	n<30	30≤n<40	n≥40
σ: known	$\bar{x} \pm z_{\alpha/2}\frac{\sigma}{\sqrt{n}}$	$\bar{x} \pm z_{\alpha/2}\frac{\sigma}{\sqrt{n}}$	?	$\bar{x} \pm z_{\alpha/2}\frac{\sigma}{\sqrt{n}}$	$\bar{x} \pm z_{\alpha/2}\frac{\sigma}{\sqrt{n}}$
σ: unknown	*To be discussed today*	$\bar{x} \pm z_{\alpha/2}\frac{s}{\sqrt{n}}$	?	*To be discussed today*	$\bar{x} \pm z_{\alpha/2}\frac{s}{\sqrt{n}}$

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Parameter of interest: true population proportion p

X: Number of successes in n trials, each trial is i.i.d. with constant p=P{success}

Random variable, $W=h(X; p) = \frac{X}{n}$ ~ Approximately Normal for large enough n.

$\hat{P} = \frac{X}{n}$

$n\hat{p} \geq 10$ and $n(1-\hat{p}) \geq 10$

$$Z = \frac{\hat{P} - p}{\sqrt{\frac{\hat{P}(1-\hat{P})}{n}}}$$

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CIs around the true population proportion p

X: Number of successes in n trials, each trial is i.i.d. with constant p=P{success}

Random variable, $Y=h(X;p)=\frac{X+2}{n+4}\sim$ Approximately Normal

$\hat{P}=\frac{X}{n}$

$\tilde{P}=\frac{X+2}{n+4}$

$\tilde{n}=n+4$

$n\hat{p}\geq 10$ and $n(1-\hat{p})\geq 10$	No condition on n or p
$\hat{p}\pm z_{\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$	$\tilde{p}\pm z_{\alpha/2}\sqrt{\frac{\tilde{p}(1-\tilde{p})}{\tilde{n}}}$

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Inferences based on two samples

$X_1,\ldots,X_m$: a random sample from a population with $\mu_1 \& \sigma_1^2$

$Y_1,\ldots,Y_n$: a random sample from a population with $\mu_2 \& \sigma_2^2$

X & Y samples are independent of one another.

Can we find a confidence interval around $\mu_1 - \mu_2$?

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A Confidence Interval for the Difference Between Two Means

$X_1,\ldots,X_m$: a random sample from a population with $\mu_1 \& \sigma_1^2$

$Y_1,\ldots, Y_n$: a random sample from a population with $\mu_2 \& \sigma_2^2$

X & Y samples are independent of one another.

When both samples are normal and the values of σ_1 and σ_2 are known :

If the two samples are independent, then a level $100(1-\alpha)\%$ CI for $\mu_X - \mu_Y$ is:

$$\bar{X} - \bar{Y} \pm z_{\alpha/2}\sqrt{\frac{\sigma_1^2}{m} + \frac{\sigma_2^2}{n}}.$$

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A Confidence Interval for the Difference Between Two Means

$X_1,\ldots,X_m$: a random sample from a population with $\mu_1 \& \sigma_1^2$

$Y_1,\ldots, Y_n$: a random sample from a population with $\mu_2 \& \sigma_2^2$

X & Y samples are independent of one another.

When the two samples are from an arbitrary distribution, the values of σ_1 and σ_2 are unknown and both m and n are large:

If the two samples are independent, then a level $100(1-\alpha)\%$ CI for $\mu_X - \mu_Y$ is:

$$\bar{X} - \bar{Y} \pm z_{\alpha/2}\sqrt{\frac{S_1^2}{m} + \frac{S_2^2}{n}}.$$

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Confidence Intervals for the Difference Between Two Proportions

Traditional approach?

New approach?

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100(1-α) Confidence Interval around $p_X - p_Y$ New Approach

$100(1-\alpha)\%$ CI for the difference $p_X - p_Y$ is

$$\tilde{p}_X - \tilde{p}_Y \pm z_{a/2}\sqrt{\frac{\tilde{p}_X(1-\tilde{p}_X)}{\tilde{n}_X} + \frac{\tilde{p}_Y(1-\tilde{p}_Y)}{\tilde{n}_Y}}$$

If the lower limit of the confidence interval is less than −1, replace it with −1.

If the upper limit of the confidence interval is greater than 1, replace it with 1.

$$\tilde{P}_X = \frac{X+1}{\tilde{n}_X}, \tilde{n}_X = n_X + 2, \tilde{P}_Y = \frac{Y+1}{\tilde{n}_Y}, \tilde{n}_Y = n_Y + 2$$

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Confidence Intervals for the Mean of a Normal r.v. with Unknown Variance:

$X_1, X_2, \ldots, X_n$: random sample with small n

Assumptions:

- $X_i \sim N(\mu, \sigma)$
- both μ and σ are unknown.

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The change in the distribution of the random variable

The random variable under consideration is still:

$$T = \frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t_{n-1}$$

How about its distribution? Still a N(0,1) r.v.?

NO!!! Why?

When n is small, S is not necessarily close to σ, thus T has more variability than Z, or in other words, T will be more spread out than the standard normal distribution.

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Definition: t Distribution

Let ν be a positive integer. Then a random variable X is said to have a t distribution with parameter ν, and the pdf of X is given by:

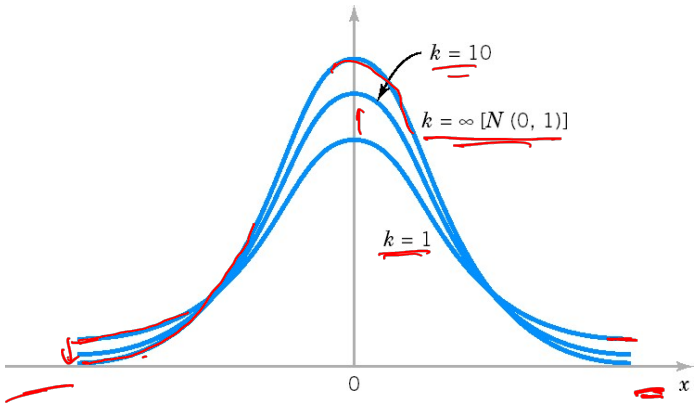
$$f(x;\nu) = \frac{\Gamma((\nu+1)/2)}{\sqrt{\pi\nu}\,\Gamma(\nu/2)} \frac{1}{((x^2/\nu)+1)^{(\nu+1)/2}} \qquad -\infty < x < \infty.$$

The parameter ν is called the number of degrees of freedom (df) of X.

Difficult to compute, so the corresponding values are tabulated (so tables are provided)

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The *t* distribution



Probability density functions of several *t* distributions.

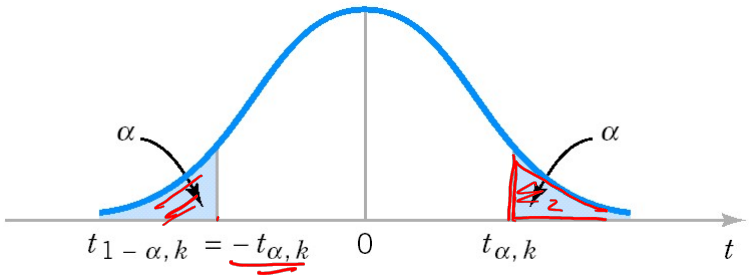
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Notation

$t_{\alpha,\nu}$ is the critical value

$t_{\alpha,\nu}$ is decreasing in both α & ν.



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t-table

$\alpha = 5\%$

$t_{9,\alpha_2} = 2.262$

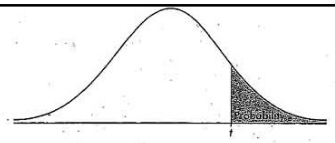


TABLE B: *t*-DISTRIBUTION CRITICAL VALUES

	Tail probability *p*											
df	.25	.20	.15	.10	.05	.025	.02	.01	.005	.0025	.001	.0005
1	1.000	1.376	1.963	3.078	6.314	12.71	15.89	31.82	63.66	127.3	318.3	636.6
2	.816	1.061	1.386	1.886	2.920	4.303	4.849	6.965	9.925	14.09	22.33	31.60
3	.765	.978	1.250	1.638	2.353	3.182	3.482	4.541	5.841	7.453	10.21	12.92
4	.741	.941	1.190	1.533	2.132	2.776	2.999	3.747	4.604	5.598	7.173	8.610
5	.727	.920	1.156	1.476	2.015	2.571	2.757	3.365	4.032	4.773	5.893	6.869
6	.718	.906	1.134	1.440	1.943	2.447	2.612	3.143	3.707	4.317	5.208	5.959
7	.711	.896	1.119	1.415	1.895	2.365	2.517	2.998	3.499	4.029	4.785	5.408
8	.706	.889	1.108	1.397	1.860	2.306	2.449	2.896	3.355	3.833	4.501	5.041
9	.703	.883	1.100	1.383	1.833	2.262	2.398	2.821	3.250	3.690	4.297	4.781
10	.700	.879	1.093	1.372	1.812	2.228	2.359	2.764	3.169	3.581	4.144	4.587
11	.697	.876	1.088	1.363	1.796	2.201	2.328	2.718	3.106	3.497	4.025	4.437
12	.695	.873	1.083	1.356	1.782	2.179	2.303	2.681	3.055	3.428	3.930	4.318
13	.694	.870	1.079	1.350	1.771	2.160	2.282	2.650	3.012	3.372	3.852	4.221
14	.692	.868	1.076	1.345	1.761	2.145	2.264	2.624	2.977	3.326	3.787	4.140
15	.691	.866	1.074	1.341	1.753	2.131	2.249	2.602	2.947	3.286	3.733	4.073
16	.690	.865	1.071	1.337	1.746	2.120	2.235	2.583	2.921	3.252	3.686	4.015
17	.689	.863	1.069	1.333	1.740	2.110	2.224	2.567	2.898	3.222	3.646	3.965
18	.688	.862	1.067	1.330	1.734	2.101	2.214	2.552	2.878	3.197	3.611	3.922
19	.688	.861	1.066	1.328	1.729	2.093	2.205	2.539	2.861	3.174	3.579	3.883
20	.687	.860	1.064	1.325	1.725	2.086	2.197	2.528	2.845	3.153	3.552	3.850
21	.686	.859	1.063	1.323	1.721	2.080	2.189	2.518	2.831	3.135	3.527	3.819
22	.686	.858	1.061	1.321	1.717	2.074	2.183	2.508	2.819	3.119	3.505	3.792
23	.685	.858	1.060	1.319	1.714	2.069	2.177	2.500	2.807	3.104	3.485	3.768
24	.685	.857	1.059	1.318	1.711	2.064	2.172	2.492	2.797	3.091	3.467	3.745
25	.684	.856	1.058	1.316	1.708	2.060	2.167	2.485	2.787	3.078	3.450	3.725
26	.684	.856	1.058	1.315	1.706	2.056	2.162	2.479	2.779	3.067	3.435	3.707
27	.684	.855	1.057	1.314	1.703	2.052	2.158	2.473	2.771	3.057	3.421	3.690
28	.683	.855	1.056	1.313	1.701	2.048	2.154	2.467	2.763	3.047	3.408	3.674
29	.683	.854	1.055	1.311	1.699	2.045	2.150	2.462	2.756	3.038	3.396	3.659
30	.683	.854	1.055	1.310	1.697	2.042	2.147	2.457	2.750	3.030	3.385	3.646
40	.681	.851	1.050	1.303	1.684	2.021	2.123	2.423	2.704	2.971	3.307	3.551
50	.679	.849	1.047	1.299	1.676	2.009	2.109	2.403	2.678	2.937	3.261	3.496
60	.679	.848	1.045	1.296	1.671	2.000	2.099	2.390	2.660	2.915	3.232	3.460
80	.678	.846	1.043	1.292	1.664	1.990	2.088	2.374	2.639	2.887	3.195	3.416
100	.677	.845	1.042	1.290	1.660	1.984	2.081	2.364	2.626	2.871	3.174	3.390
1000	.675	.842	1.037	1.282	1.646	1.962	2.056	2.330	2.581	2.813	3.098	3.300
∞	.674	.841	1.036	1.282	1.645	1.960	2.054	2.326	2.576	2.807	3.091	3.291

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The change in the distribution of the random variable

The random variable under consideration is still:

$$T = \frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t_{n-1}$$

CI?

How about its distribution? Still a N(0,1) r.v.?

NO!!! Why?

When n is small, S is not necessarily close to σ, thus T has more variability than Z, or in other words T will be more spread out than standard normal dist.

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$$T = \frac{\bar{X} - \mu}{S/\sqrt{n}}$$

Proposition

Let $\bar{x}$ and s be the sample mean and sample standard deviation computed from the results of a random sample of n from a normal population with mean μ. Then

$$\left(\bar{x} - t_{n-1,\alpha/2}\frac{s}{\sqrt{n}}, \bar{x} + t_{n-1,\alpha/2}\frac{s}{\sqrt{n}}\right)$$

is a confidence interval for μ with confidence level 100(1-α)%.

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Example: A cargo company

A cargo company aims to have a mean delivery time of 5 hours for local deliveries.

They believe that delivery times are normal with σ is not known.

They recorded 10 delivery times as:

7.3, 3.3, 4.7, 6.8, 10.1, 5.7, 6.4, 4.2, 3.1, 8.9

95%CI on the true mean delivery time?

$\bar{x} = 6.05$
$s = 2.32$

95% CI with known $\sigma = 2.4$: (4.56, 7.54), with width = 2.98

95% CI with unknown σ estimated by s = 2.32: (4.39, 7.71), with width = 3.32

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When to use t-distribution?

The data should:

- Have no outliers,
- Not display a strong asymmetry.

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Main idea:

Collect a random sample

Determine the parameter of interest, say θ

Find a random variable, say W, that involves parameter θ and whose distribution you can specify

Use the distribution of W to find random lower and upper bounds on θ

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Parameter of interest: true mean, μ

- $X_1, \ldots, X_n$: A random sample with $E[X] = \mu$
- We are looking for a random variable $Y = h(X_1, \ldots, X_n; \mu)$ whose distribution can be specified.

	X is normal		X is not known to be normal		
	n<40	n≥40	n<30	30≤n<40	n≥40
σ: known	$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$	$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$	?	$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$	$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$
σ: unknown	$T = \frac{\bar{X} - \mu}{S/\sqrt{n}}$	$Z = \frac{\bar{X} - \mu}{S/\sqrt{n}}$	?	$T = \frac{\bar{X} - \mu}{S/\sqrt{n}}$	$Z = \frac{\bar{X} - \mu}{S/\sqrt{n}}$

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CIs around the true mean, μ

- $X_1, \ldots, X_n$: A random sample with $E[X] = \mu$
- We are looking for a random variable $Y=h(X_1,\ldots,X_n; \mu)$ whose distribution can be specified.

	X is normal		X is not normal		
	n<40	n≥40	n<30	30≤n<40	n≥40
σ: known	$\bar{x} \pm z_{\alpha/2}\frac{\sigma}{\sqrt{n}}$	$\bar{x} \pm z_{\alpha/2}\frac{\sigma}{\sqrt{n}}$	?	$\bar{x} \pm z_{\alpha/2}\frac{\sigma}{\sqrt{n}}$	$\bar{x} \pm z_{\alpha/2}\frac{\sigma}{\sqrt{n}}$
σ: unknown	$\bar{x} \pm t_{\alpha/2}\frac{s}{\sqrt{n}}$	$\bar{x} \pm z_{\alpha/2}\frac{s}{\sqrt{n}}$	?	$\bar{x} \pm t_{\alpha/2}\frac{s}{\sqrt{n}}$	$\bar{x} \pm z_{\alpha/2}\frac{s}{\sqrt{n}}$

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Parameter of interest: true population proportion p

X: Number of successes in n trials, each trial is i.i.d. with constant p=P{success}

We are looking for a random variable , $Y=h(X; \theta)$ whose distribution can be specified.

$$\hat{P} = \frac{X}{n}$$

$n\hat{p}\geq 10$ and $n(1-\hat{p})\geq 10$

$$Z = \frac{\hat{P} - p}{\sqrt{\frac{\hat{P}(1-\hat{P})}{n}}}$$

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CIs around the true population proportion p

X: Number of successes in n trials, each trial is i.i.d. with constant p=P{success}

We are looking for a random variable , $Y=h(X;\theta)$ whose distribution can be specified.

$\hat{P} = \frac{X}{n}$

$\tilde{P} = \frac{X+2}{n+4}$

$\tilde{n} = n+4$

$n\hat{p}\geq 10$ and $n(1-\hat{p})\geq 10$	**No condition on n or p**
$\hat{p} \pm z_{\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$	$\tilde{p} \pm z_{\alpha/2}\sqrt{\frac{\tilde{p}(1-\tilde{p})}{\tilde{n}}}$

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Small and Normal Sample Confidence Intervals for the Difference Between Two Means

$X_1,\ldots,X_m$: a random sample from a population with $\mu_1 \& \sigma_1^2$

$Y_1,\ldots, Y_n$: a random sample from a population with $\mu_2 \& \sigma_2^2$

X & Y samples are independent of one another.

When the two samples are from ~~an arbitrary~~ a normal distribution, the values of σ_1 and σ_2 are unknown.

$$\frac{\bar{X}-\bar{Y}-(\mu_1-\mu_2)}{\sqrt{\frac{S_1^2}{m}+\frac{S_2^2}{n}}} \sim t_{df}$$

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Small-Sample Confidence Intervals for the Difference Between Two Means

$100(1-\alpha)\%$ CI for the difference $\mu_X - \mu_Y$ is

$$\bar{X} - \bar{Y} \pm t_{v,\alpha/2}\sqrt{\frac{s_X^2}{n_X} + \frac{s_Y^2}{n_Y}}.$$

$$v = \frac{\left(\frac{s_X^2}{n_X} + \frac{s_Y^2}{n_Y}\right)^2}{\frac{\left(\frac{s_X^2}{n_X}\right)^2}{n_X - 1} + \frac{\left(\frac{s_Y^2}{n_Y}\right)^2}{n_Y - 1}}$$

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When the Populations Have Equal Variances

Suppose we have the same set-up as before, but the populations are known to have nearly the same variance.

Then a $100(1-\alpha)\%$ CI for $\mu_X - \mu_Y$ is

$$\bar{X} - \bar{Y} \pm t_{n_X+n_Y-2,\alpha/2} \cdot s_p\sqrt{\frac{1}{n_X} + \frac{1}{n_Y}}.$$

pooled t-test

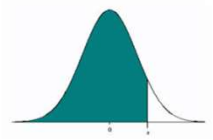
The quantity s_p is the pooled standard deviation, given by

$$s_p = \sqrt{\frac{(n_X - 1)s_X^2 + (n_Y - 1)s_Y^2}{n_X + n_Y - 2}}.$$

The best practice is to assume the variances are unequal unless it is quite certain that they are equal.

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Table of Standard Normal Probabilities for Positive Z-scores



z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998

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